

# SynPO: Synergizing Descriptiveness and Preference Optimization for Video Detailed Captioning

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**Abstract**—Fine-grained video captioning aims to generate detailed, temporally coherent descriptions of video content. However, existing methods struggle to capture subtle video dynamics and rich detailed information. In this paper, we leverage preference learning to enhance the performance of vision-language models (VLM) in fine-grained video captioning, while mitigating several limitations inherent to Direct Preference Optimization (DPO). First, we propose a pipeline for constructing preference pairs that leverages the intrinsic properties of VLMs along with partial assistance from large language models, achieving an balance between cost and data quality. Then, we propose Synergistic Preference Optimization (SynPO), a novel optimization method offering significant advantages over DPO and its variants. SynPO prevents negative pReferences from dominating the training, explicitly preserves the model's language capability to avoid deviation of the optimization objective, thus obtains high-quality captions and improves training efficiency by eliminating the need for the reference model. We extensively evaluate our proposed data construction pipeline across three models: AuroraCap, LLaVA1.6-7B-Video and InterVL2-8B. Results demonstrate that our method improve performance in fine-grained video captioning significantly and consistently. Source code is available at <https://github.com/longmalongma/SynPO>

**Index Terms**—Vision-language model, preference optimization, video captioning.

## I. INTRODUCTION

FINE-GRAINED video captioning aims to generate detailed and coherent textual descriptions that precisely capture video contents. This task necessitates the recognition of salient actions and objects, while also modeling fine-grained visual features and temporal dynamics. Recent studies [1], [2], [3], [4] have primarily employed Vision-Language Models (VLMs) for video captioning. These methods [5], [6], [7] typically utilize pre-trained vision encoders and Large Language

Models (LLMs), with a connector module linking them. By training on video-text pairs, these models aim to align visual and textual representations, thereby enhancing their ability to understand and describe video content [8] effectively.

Direct Preference Optimization (DPO) [9] is a fine-tuning method that aligns models with stipulated preferences using high-quality preference pairs. Recent work has successfully adapted DPO and its variants to video understanding tasks, significantly improving model performance [10], [11]. Thus, integrating DPO into fine-grained video captioning to enhance the model's ability to capture temporal dynamics and detailed descriptions present a promising direction. However, two critical challenges currently degrade its performance in fine-grained video captioning: (1) the scarcity of high-quality video-text alignment pairs, which are essential for preference learning; (2) DPO typically suffers from the simultaneous decrease in both positive and negative reward values [12], [13], leading to a potential objective optimization deviation from focusing on generation quality to merely discriminating between preferences [14], as shown in Figure 1 (middle).

Recently, the proposed VDC benchmark [15] is well-curated, but limited in scale and partially reliant on manual annotations. Other video captioning datasets, such as MSRVT [16], VATEX [17], MSVC [1], etc., typically provide overly brief captions, falling short in fine-grained video captioning. Besides, these datasets lack preference pairs, and thus cannot be prepared for DPO. To construct preference pairs, many existing methods [18], [19] rely on a stronger VLM to score multiple outputs from the same prompt. While straightforward, this approach is impractical: small teams face prohibitive API costs, while developers of powerful models often lack access to a stronger scoring model. Some studies attempt to circumvent this limitation by generating negative preferences, such as through atypical item substitution [20] or temporal perturbation [10]. However, these methods primarily focus on negative samples and fail to produce higher-quality positive preferences.

We present an automated pipeline for constructing high-quality preference pairs for fine-grained video captioning. Given the same input, we generate multiple alternative outputs using a VLM. These candidates are then scored leveraging intrinsic properties of the VLM itself, such as its self-consistency [21] and the enhanced ability to capture details in short videos, with limited assistance from an LLM. The top and bottom scores are selected as positive and negative

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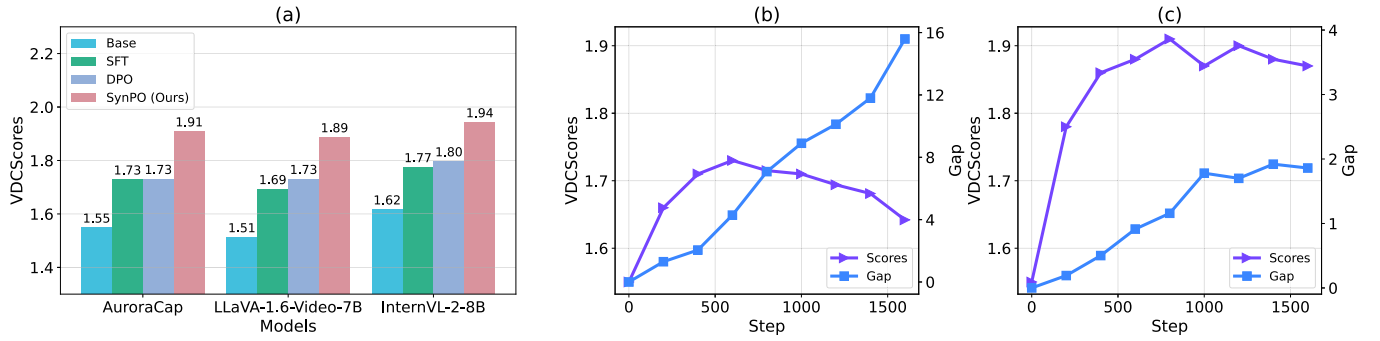


Fig. 1. a: SynPO significantly outperforms other methods in different models on VDC benchmark [15]. Here the VDC Score is employed to evaluate captions generated by models. b: Evolution of the VDC score and the gap between positive and negative rewards during DPO fine-tuning, illustrating while the reward gap increases continuously during DPO fine-tuning, the quality of captions generated by model initially improves and subsequently declines. It shows that DPO excessively optimizes the reward gap. c: SynPO mitigates degradation successfully and resolves the issue of optimization objectives shifting from generating high-quality captions to ranking differentiation. Its performance significantly outperforms that of DPO.

preferences, respectively, forming our constructed preference dataset. Compared to existing approaches, our method achieves an balance between cost efficiency and high-quality preference pair construction.

We propose an improved optimization method for DPO, termed Synergistic Preference Optimization (SynPO). Our SynPO features three critical advantages: (1) It reformulates the reward gap computation to prevent the influence of negative preferences from dominating the optimization process, thereby fundamentally addressing the issue of simultaneous decreases in both positive and negative reward values; (2) It introduces an additional reward term in the loss function that explicitly encourages language capability, helping to maintain the model's generative performance and prevent objective deviation during optimization, shown in Figure 1 (c); (3) It eliminates the need for a reference model during training, resulting in an approximately 20% improvement in training efficiency.

We apply our proposed data generation pipeline to construct preference datasets for three video-capable models: AuroraCap [15], LLaVA1.6-7B-Video [22], and InterVL2-8B [23]. These datasets are then used to optimize the respective models via SynPO and DPO. As shown in Figure 1 (left), the results show that models fine-tuned on the generated preference data achieve significantly improve performance in fine-grained video captioning compared to their original versions, demonstrating the efficacy of the proposed data generation pipeline. Furthermore, SynPO consistently outperforms DPO and its variants in terms of alignment effectiveness, validating its superiority in preference optimization. To assess generalization, we further adapt SynPO to natural language processing tasks by applying it to Mistral-7B [24] (Base and Instruct) and Llama3-8B [25] (Base and Instruct). The consistent performance gains across these models confirm that SynPO is not only effective but also broadly applicable across different architectures and task domains.

The contributions of this paper are three-fold: 1) We propose a novel pipeline that automatically generates high-quality preference pairs for fine-grained video captioning by leveraging a VLM's intrinsic self-consistency and detail-capturing ability; 2) We introduce SynPO, an improved DPO

method that preserves the contrastive optimization mechanism to enhance preference discrimination capability while incorporating an additional language reward term. This term ensures that each token contributes uniformly to the gradient computation, thereby delivering a stable and smooth language modeling signal at the sentence level. The two components operate synergistically: the contrastive term amplifies critical distinctions to strengthen preference alignment, whereas the language reward term mitigates excessive deviations induced by low-probability tokens, thereby preserving the model's overall linguistic quality. 3) Extensive experiments on video captioning demonstrate SynPO's superiority over several DPO variants. Our approach also achieves superior results on NLP tasks, verifying its effectiveness across domains.

## II. RELATED WORK

### A. Video Captioning

Video captioning aims to generate natural language descriptions of video content, integrating visual understanding with linguistic expression. Early studies [26], [27] employed template-based approaches, assembling subject-verb-object triplets into predefined structures, yielding constrained captions. For instance, Kojima et al. [27] developed action hierarchies for activity descriptions, while Rohrbach et al. [28] utilized Conditional Random Fields to capture video segment relationships. The emergence of Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks facilitated more dynamic captions. Studies such as [29] and [30] modeled captioning as machine translation, adopting an encoder-decoder framework to encode video features and decode them into text. However, RNNs faced challenges in capturing long-range dependencies, limiting performance on complex videos. Transformer models addressed these limitations through self-attention mechanisms [31], yielding notable progress. CLIP4Caption [32] harnesses CLIP's vision-language pre-training to improve video-text alignment, producing accurate captions on MSR-VTT. SwinBERT [33] pioneers end-to-end training with dense frame sampling and sparse attention, achieving state-of-the-art performance across datasets like ActivityNet Captions. ShareGPT4Video [34]

enhances model generalization with 40K GPT4V-annotated captions, supporting precise descriptions for diverse videos, complementing advances in dense video captioning [35]. These advancements in modeling and data quality significantly elevate video captioning capabilities.

### B. Vision-Language Models

Recent advancements in LLM [36], [37], [38], [39] have spurred the development of VLM [1], [2], [3], [4]. These models build upon foundational efforts in aligning vision and language modalities, with CLIP [40] serving as a seminal example. CLIP leverages contrastive learning on extensive image-text pair datasets to achieve robust alignment between visual and textual representations. Early VLM adapted frameworks originally designed for image-language tasks, such as Flamingo [41] and BLIP-2 [5], to process video inputs. These approaches typically flattened the spatio-temporal features of videos into one-dimensional sequences, which were subsequently fed into LLMs. However, this method proved inadequate in capturing the inherent temporal dynamics of video data, thus constraining its performance in tasks requiring sequential comprehension. Later models, such as VideoChat [6] and VideoLLaMA [7], addressed this limitation by modifying the LLM architecture to better accommodate the sequential nature of video content. Similarly, building on LLaVA [42], Video-ChatGPT [43] averages the frame-level features across both spatial and temporal dimensions to create a video-level representation. ChatVideo [8] treats short, coherent video segments, known as tracklets, as the fundamental video elements. It relies on several video foundation models to annotate these tracklets with textual descriptions, and then uses ChatGPT [44] to perform various video understanding tasks.

### C. Reinforcement Learning

1) *Reinforcement Learning From Human Feedback*: Reinforcement Learning from Human Feedback (RLHF) [45] aligns LLMs with human preferences [39], [46], [47] through supervised fine-tuning [48], reward modeling [49], and policy optimization, improving instruction-following [39] and safety [50]. It first trains a reward model on the preference dataset:

$$\mathcal{L}_R(r_\phi, \mathcal{D}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} [\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))], \quad (1)$$

Then the learned reward model is used to provide feedback to the language model. The optimization is formulated as:

$$\begin{aligned} \max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(y|x)} [r_\phi(x, y)] \\ - \beta \mathbb{D}_{\text{KL}}[\pi_\theta(y|x) \parallel \pi_{\text{ref}}(y|x)], \end{aligned} \quad (2)$$

DPO provides a streamlined alternative to RLHF by directly optimizing the language model using preference data without explicit reward modeling or reinforcement learning steps. DPO reformulates the RLHF optimization objective into a supervised learning problem, eliminating the need for separate reward model training and policy optimization. Instead of using a reward model to guide policy updates, DPO directly maps human preferences to policy updates by minimizing a loss that aligns the model's probability ratios with preference

rankings. By directly incorporating preference data into the policy optimization process, DPO achieves comparable alignment performance to RLHF while avoiding common RLHF challenges such as reward model bias, training instability, and the computational overhead of reinforcement learning loops.

2) *Group Relative Policy Optimization*: Group relative policy optimization (GRPO) [51] is a new online training paradigm designed specifically for reasoning tasks. In each training iteration, the LLM generates a group of responses to a given query. These responses are subsequently evaluated using a predefined rule-based reward function, and the resulting rewards are standardized prior to model updates via policy optimization [52]. This approach exploits group-level feedback signals to guide the model toward more accurate and coherent reasoning behaviors.

## III. CONSTRUCTING LONG VIDEO CAPTION PREFERENCE PAIRS

### A. Enhancing Model Output

The first part of our work is to build a pipeline to construct high-quality preference dataset for video captioning. To address the challenges of hallucination and generic outputs in fine-grained video captioning, we incorporate contrastive decoding and a self-retrospective strategy. These methods target complementary aspects: contrastive decoding enhances factual consistency by reducing hallucinations, while the self-retrospective approach improves the model's ability to capture and articulate detailed visual content.

Contrastive decoding [53] was initially proposed to reduce object hallucinations and subsequently improved [54] by introducing a more efficient variant that contrasts logits from sparse frame samples with those from full sequences. In this work, the improved contrastive decoding method is adopted to suppress overconfidence in noisy or irrelevant features, thereby improving factual consistency and detail accuracy in generated captions.

The self-retrospective strategy, introduced by [55], operates independently of the decoding process. It enhances comprehension through iterative refinement by feeding the model's own outputs back into the input prompt. This method draws inspiration from human retrospective thinking [56], allowing the system to adjust its predictions based on previous outputs. We adapt this approach to video captioning by incorporating the model's initial draft as contextual input for subsequent refinement steps, enabling the generation of richer and more coherent descriptions.

These two strategies are integrated by applying contrastive decoding at both stages of the self-retrospective process. Specifically, contrastive decoding is utilized when generating the initial caption and again during the refinement step. This ensures that each iteration benefits from a reduction in hallucinations and an improvement in fidelity, while also leveraging the iterative refinement capabilities of the self-retrospective strategy to enhance descriptive richness and linguistic fluency.

### B. Dataset Construction Pipeline

We propose an automated pipeline for constructing high-quality preference pairs specifically designed for fine-grained

video captioning. The overall framework is designed to address the limitations of existing datasets, such as limited scale, insufficient detail in captions, and the lack of human-like preference annotations [1], [15], [17]. Our method leverages both the intrinsic properties of VLMs and the reasoning capabilities of LLMs to generate diverse and reliable preference pairs without dependence on costly multimodal scorers.

The core strategy involves generating multiple candidate captions for each video using a single VLM under the same prompt. These candidates are then scored using a novel three criterion evaluation framework that combines factuality, modality correctness, and self-consistency [21]. Based on the aggregated scores across all three criteria, the captions with the highest and lowest total scores are selected as positive and negative preferences, respectively, forming our final dataset. In the following, we describe each scoring criterion in detail.

1) *Criterion 1: Factuality Through Temporal Decomposition*: Due to input token length constraints in most VLMs, processing long videos directly often leads to loss of detail and increased hallucination [10]. To mitigate this, we segment each video into short clips and independently process them using the VLM to generate clip-level captions. These are concatenated into a reference set used to evaluate the factuality of the full-video caption generated from the entire input. An LLM then assesses the consistency between the full-video caption and the reference set, focusing particularly on factual alignment. This design improves detail preservation while reducing hallucinatory content. Scores range from 0 to 5.

2) *Criterion 2: Instruction Fidelity, Linguistic Fluency and Objectivity*: Video captions generated by the VLM are assessed by an LLM according to the following criteria: (1) Instruction fidelity: Whether the caption meets the requirements of the corresponding prompt; (2) Linguistic fluency: Whether the description is natural and coherent, using language appropriate for describing a video (e.g., avoid calling a video an “image”); (3) Objectivity: Minimizing subjective or illogical content. Each caption receives an overall score between 0 and 5, ensuring semantically accurate and linguistically well-formed outputs.

3) *Criterion 3: Self-Consistency Through Multi-Sample Analysis*: Inspired by self-consistency methods in NLP [57], [58], [59], we apply it to video captioning by assessing the stability of key entities, actions, and temporal dynamics across multiple generations. Specifically, the VLM generates  $n$  diverse captions via high-temperature sampling. An LLM analyzes their similarity, rewarding consistent patterns and penalizing outliers through a majority voting mechanism. Given its narrower discriminative capacity compared to the other two criteria, this metric uses a 0 to 3 scoring range.

#### IV. SOLUTION: SYNERGIZING DESCRIPTIVENESS AND PREFERENCE OPTIMIZATION

##### A. Preliminaries

The substance of DPO is that there exists an implicit reward function  $r^*(x, y)$  from which human preferences are generated.

An ideal model  $\pi^*$  would assign probabilities according to this reward function:

$$\pi^*(y|x) \propto \pi_{\text{ref}}(y|x) \exp(\beta r^*(x, y)) \quad (3)$$

where  $\pi_{\text{ref}}(y|x)$  is the probability from the reference model which is typically a model fine-tuned via Supervised Fine-Tuning (SFT).  $\beta$  is a hyperparameter that controls the extent to which the reward function influences the probability distribution.

The objective of DPO is to find a policy  $\pi_\theta$  with model parameters  $\theta$  that best explains the observed preference data. This is achieved by maximizing the following objective function:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim D} \left[ \log \sigma \left( \beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right] \quad (4)$$

where  $D$  is the preference dataset, consisting of triplets  $(x, y_w, y_l)$ .  $y_w$  is the preferred response, and  $y_l$  is the less preferred response.  $\sigma(z) = 1/(1 + e^{-z})$  is the sigmoid function.  $\beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$  can be viewed as the implicit reward  $\hat{r}_\theta(x, y)$  assigned by the model  $\pi_\theta$  relative to the reference model  $\pi_{\text{ref}}$ .

DPOP [12] is a DPO variant which demonstrates that although the standard DPO loss function is designed to increase the relative probability of the preferred response  $y_w$  over the less preferred response  $y_l$  (i.e., enlarging  $\frac{\pi_\theta(y_w|x)}{\pi_\theta(y_l|x)}$ ), it may inadvertently lead the model to decrease  $\log \pi_\theta(y_w|x)$ , the absolute likelihood of the preferred response.

SimPO [60] is another variant of DPO which aims to eliminate the additional computational overhead in DPO and introduces length normalization to the reward function, addressing the limitation of DPO in not accounting for sequence length. In addition, an important improvement of SimPO is the removal of the KL divergence term in DPO. It has been shown that even without the KL divergence constraint, the model can still achieve effective learning while maintaining generalization by adjusting other factors in preference learning, such as the learning rate and dataset. Its objective function is:

$$\mathcal{L}_{\text{SimPO}}(\pi_\theta) = -\mathbb{E}_{(x, y_w, y_l) \sim D} \left[ \log \sigma \left( \frac{\beta}{|y_w|} \log \pi_\theta(y_w | x) - \frac{\beta}{|y_l|} \log \pi_\theta(y_l | x) - \gamma \right) \right] \quad (5)$$

##### B. DPO: Scope for Further Development

Despite the advantages of DPO, there remains potential for improvement. Our work refines DPO primarily based on two key observations: negative rewards tend to dominate during training, and the optimization objective deviates from the goal of generating high-quality sentences.

1) *Negative Preference Dominates Training Process*: As shown in Figure 3, during DPO training, both positive and negative rewards decrease, with the negative reward decreasing more rapidly. This indicates that the model is encouraged more to reduce negative reward, rather than both increasing positive

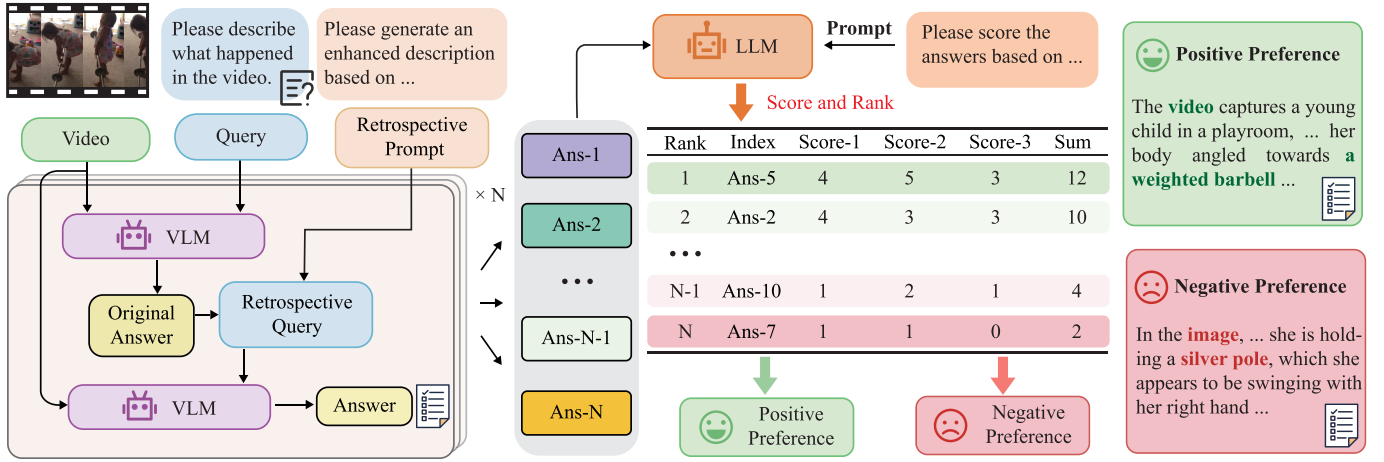


Fig. 2. Overview of dataset construction pipeline. A VLM first generates multiple candidate captions for each video with the self-retrospective strategy. Then the candidate captions are scored by an LLM based on three criteria (i.e., factuality, linguistic fluency, and self-consistency) to select positive and negative preferences.

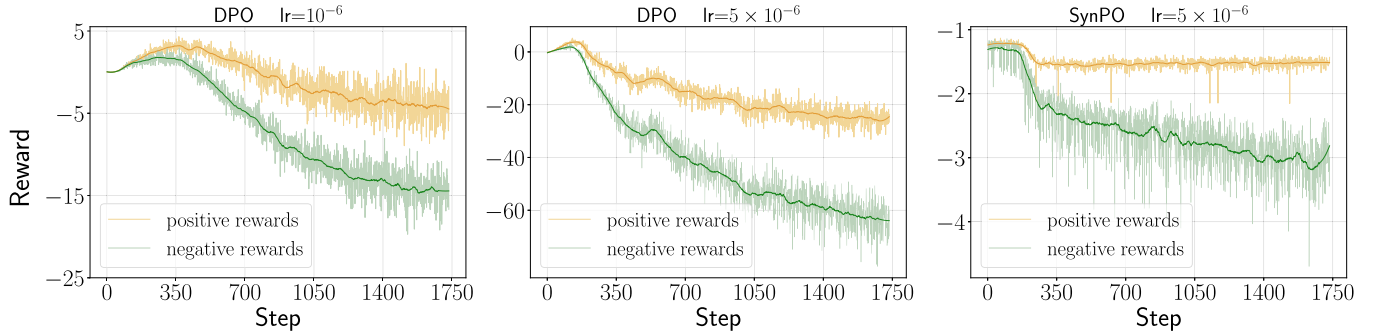


Fig. 3. The evolution of positive and negative rewards during training of DPO and SynPO. DPO training undergoes simultaneous decreases in both rewards, with negative preferences dominating the optimization process. Conversely, SynPO mitigates this problem, demonstrating improved performance and stability.

reward and reducing negative rewards. More precisely, the negative reward term exerts greater influence on the parameter updates.

From the perspective of the gradient of DPO with respect to the model parameters  $\theta$ , the inherent symmetry of the formulation suggests that positive and negative rewards exert seemingly identical influences on the parameters. To further investigate this issue, we attempt to decompose the gradient from another perspective, thereby uncovering the essence of the problem. We reformulate the gradient as:

$$\frac{\partial \mathcal{L}_{DPO}}{\partial \pi_{\theta}} = \frac{\partial \mathcal{L}_{DPO}}{\partial r(x, y_w)} \frac{\partial r(x, y_w)}{\partial \pi_{\theta}} + \frac{\partial \mathcal{L}_{DPO}}{\partial r(x, y_l)} \frac{\partial r(x, y_l)}{\partial \pi_{\theta}} \quad (6)$$

Eq. 6 indicates that the gradient of  $\mathcal{L}_{DPO}$  with respect to the model parameters can be divided into two parts. We attempt to show that the two have different impact on the overall gradient. The gradient of  $\mathcal{L}_{DPO}$  with respect to positive reward is:

$$\frac{\partial \mathcal{L}_{DPO}}{\partial r(x, y_w)} = -\mathbb{E}_{(x, y_w, y_l) \sim D} \left[ \frac{\partial \mathcal{L}_{DPO}}{\partial \sigma(\beta(r(x, y_w) - r(x, y_l)))} \cdot \frac{\partial \sigma(\beta(r(x, y_w) - r(x, y_l)))}{\partial \beta(r(x, y_w) - r(x, y_l))} \cdot \frac{\partial \beta(r(x, y_w) - r(x, y_l))}{\partial r(x, y_w)} \right]$$

$$\begin{aligned} &= -\mathbb{E}_{(x, y_w, y_l) \sim D} \left[ \frac{1}{\sigma(\beta(r(x, y_w) - r(x, y_l)))} \cdot \sigma(\beta(r(x, y_w) - r(x, y_l))) \cdot (1 - \sigma(\beta(r(x, y_w) - r(x, y_l)))) \cdot \beta \right] \\ &= -\mathbb{E}_{(x, y_w, y_l) \sim D} \beta (1 - \sigma(\beta(r(x, y_w) - r(x, y_l)))) \quad (7) \end{aligned}$$

To quantify the influence of each gradient on the model parameters, we use the normalized Frobenius norm to measure the magnitude of each gradient. The two terms,  $\frac{\partial \mathcal{L}_{DPO}}{\partial r(x, y_w)}$  and  $\frac{\partial \mathcal{L}_{DPO}}{\partial r(x, y_l)}$ , differ only by a sign, their update strength on the gradient is identical. More precisely, Formula 6 indicates that during backpropagation, gradients propagate from the loss function to the reward values associated with the preferred response  $y_w$  and the rejected response  $y_l$ , respectively, and subsequently converge to update the model parameters. The gradient computation comprises two distinct phases: specifically, the contribution of the loss function to the model parameters via the rejected response reward equals the product of the gradient of the loss with respect to that reward and the gradient of the reward with respect to the model parameters. The derivation presented in Formula 7 establishes that the

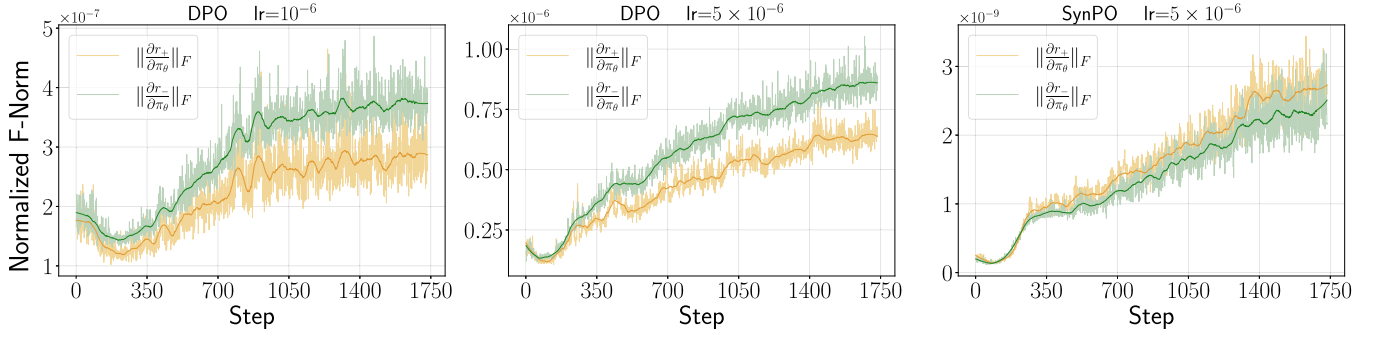


Fig. 4. The evolution of the Normalized Frobenius Norm ( $\|A\|_F/mn$  for  $A \in \mathbb{R}^{m \times n}$ , normalized by the number of matrix elements) with respect to training steps. Here the Normalized Frobenius Norm is employed to measure the strength of model parameter update in each training step.

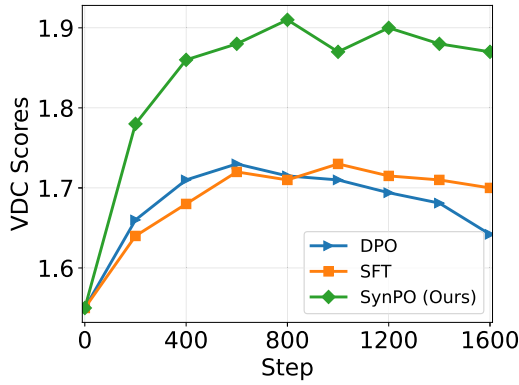


Fig. 5. Caption quality under different fine-tuning methods. VDC Score is employed as evaluation metric.

gradients of the loss function with respect to the rewards for the preferred and rejected responses are equal in magnitude but opposite in sign. Thus, our work further investigates the impact of the other two terms. As shown in Figure 4, the Normalized Frobenius Norms of  $\frac{\partial r(x, y_w)}{\partial \pi_\theta}$  is greater. This indicates that  $r(x, y_l)$  exerts a stronger influence on the model parameter updates, thereby demonstrating that negative reward dominates the optimization process.

2) *Deviation in Optimization Objective*: As mentioned earlier, the DPO objective function optimizes the reward gap. However, we observe that as the reward gap increases, the model's language capability does not show consistent improvement. As shown in Figure 1, during DPO training, the reward gap keeps increasing, fitting the increased score on the preference task. However, the model's score shows a trend of first increasing and then decreasing on the language task. This shows that the enhancement of the ranking ability through DPO does not always translate into language capability. More precisely, as shown in Figure 5, DPO-tuned model exhibits initial rapid improvement and subsequent performance drop, suggesting that DPO may inadvertently weaken the quality of generated captions.

We offer a token-level perspective to interpret this problem. Consider the gradient of the DPO loss with respect to the model output logits. For a token  $y_{w,t}$  in the preferred sequence

$y_w$ , its gradient contribution can be approximated as:

$$\frac{\partial L_{\text{DPO}}}{\partial \text{logit}(y_{w,t})} \propto (1 - \sigma(\Delta r)) \left( \frac{1}{\pi_\theta(y_{w,t} | x, y_{w,1:t-1})} - \frac{1}{\pi_{\text{ref}}(y_{w,t} | x, y_{w,1:t-1})} \right) \quad (8)$$

where

$$\Delta r = r(x, y_w) - r(x, y_l) \quad (9)$$

For a token  $y_{l,t}$  in the less-preferred sequence  $y_l$ , its gradient contribution can be approximated as:

$$\frac{\partial L_{\text{DPO}}}{\partial \text{logit}(y_{l,t})} \propto -(1 - \sigma(\Delta r)) \left( \frac{1}{\pi_\theta(y_{l,t} | x, y_{l,1:t-1})} - \frac{1}{\pi_{\text{ref}}(y_{l,t} | x, y_{l,1:t-1})} \right) \quad (10)$$

Although these gradients aim to increase the relative reward of  $y_w$  and decrease that of  $y_l$ , when  $y_w$  and  $y_l$  are highly similar, these updates can interact in complex ways, potentially leading to a decrease in the overall probability of  $y_w$ . DPO may end up focusing primarily on the logits of a few discriminative tokens.

### C. SynPO: Synergistic Preference Optimization

To address the aforementioned limitations, we propose a novel DPO variant named **Synergistic Preference Optimization** (SynPO), which enhances preference alignment while preserving strong language modeling capabilities. Our objective function is formulated as:

$$\begin{aligned} \mathcal{L}_{\text{SynPO}} = & -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[ \sigma \left( \right. \right. \\ & \alpha \cdot \exp \left( \frac{1}{|y_w|} \sum_{t=1}^{|y_w|} \log \pi_\theta(y_{w,t} | x, y_{w,<t}) \right) \\ & - \alpha \cdot \exp \left( \frac{1}{|y_l|} \sum_{t=1}^{|y_l|} \log \pi_\theta(y_{l,t} | x, y_{l,<t}) \right) \\ & \left. \left. + \beta \cdot \frac{1}{|y_w|} \sum_{t=1}^{|y_w|} \pi_\theta(y_{w,t} | x, y_{w,<t}) \right) \right] \quad (11) \end{aligned}$$

where  $\alpha$  and  $\beta$  are hyperparameters,  $y_w$  and  $y_l$  denote the positive and negative preferences respectively.  $|y|$  denotes the

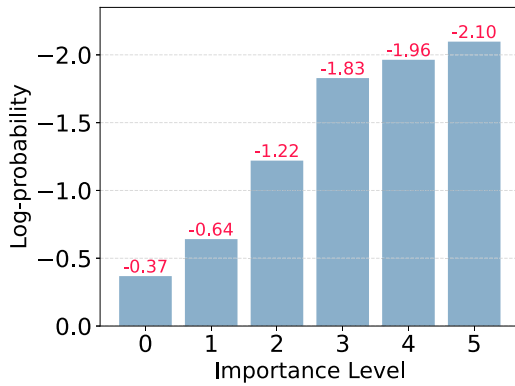


Fig. 6. The log-probability of tokens. X-axis represents the semantical importance scores of specific tokens given by LLM.

sequence length of the preferences.  $\pi_\theta(y_t|x, y_{<t})$  represents the probability of the current model output for the  $t$ -th token.

The SynPO objective primarily consists of two components: a contrastive optimization term and a language reward term. To reduce misunderstanding, we begin with a theoretical analysis of the difference between the language reward term and the positive preference component in the contrastive optimization term. In the following, we present an analysis of the gradient expressions of these two terms to clarify their distinctions. For notational convenience, we temporarily denote  $p_t$  as  $\pi_\theta(y_t|x, y_{<t})$ .

The gradient of one component in the contrastive optimization term with respect to  $p_t$  is given by:

$$\nabla_p \exp \left( \frac{1}{|y|} \sum_{i=1}^{|y|} \log p_i \right) = \frac{1}{|y|} \cdot \exp \left( \frac{1}{|y|} \sum_{i=1}^{|y|} \log p_i \right) \cdot \left( \frac{1}{p_1}, \frac{1}{p_2}, \dots, \frac{1}{p_{|y|}} \right)^T \quad (12)$$

The gradient of the language reward term with respect to  $p_t$  is given by:

$$\nabla_p \left( \frac{1}{|y|} \sum_{i=1}^{|y|} p_i \right) = \left( \frac{1}{|y|}, \frac{1}{|y|}, \dots, \frac{1}{|y|} \right)^T \quad (13)$$

As can be observed from Eq. 12,  $\exp \left( \frac{1}{|y|} \sum_{i=1}^{|y|} \log p_i \right)$  is more sensitive to smaller values of  $p_t$ . In other words, at the sentence-level, tokens with lower output probabilities from the model have a greater impact on parameter updates. In contrast, as shown in Eq. 13,  $\left( \frac{1}{|y|} \sum_{i=1}^{|y|} p_i \right)$  does not exhibit this behavior. Regardless of their individual probability values, each token contributes equally to the parameter updates.

1) *Observations About Token Probabilities:* As illustrated in Figure 6, the model tends to output lower probability values for tokens that are more critical and have a greater influence on the semantic trajectory. This observation aligns with empirical evidence: for easily predictable tokens, such as conjunctions and the latter parts of complete phrases, the model is often highly confident and outputs relatively higher probability values. Conversely, for tokens that are less straightforward to predict, such as key calculations in mathematical solutions, the

model exhibits less confidence. It considers several alternative tokens as potentially correct, resulting in comparatively lower probability values for these tokens.

2) *Contrastive Optimization Term:* Based on observations about token probabilities, regarding the contrastive optimization term, although we have identified that the logarithmic transformation can lead to an excessive influence of negative preferences over positive ones, we do not simply eliminate this logarithmic transformation. Instead, we apply an additional exponential transformation to the entire log-mean function. This approach not only mitigates the adverse effects caused by the logarithmic transformation but also preserves its beneficial property of amplifying the gradient contributions of tokens with lower probabilities. This characteristic aligns well with the objectives of contrastive learning, as it enables the optimization process to focus more on the critical tokens that contribute to the distinction between positive and negative preferences.

3) *Language Reward Term:* Compared to the DPO objective, we introduce an additional reward term specifically designed to encourage the model to maintain its language generation capabilities, thereby preventing the optimization objective from drifting from that of a generative language model toward a mere ranking model. This auxiliary term employs an arithmetic mean. As our previous analysis indicates, the arithmetic mean ensures that each token contributes equally to the update, regardless of its individual probability. This uniform contribution is precisely the desired property for this component. From the perspective of preserving language proficiency, every token should be treated equally, as they collectively contribute to the fluency, grammatical correctness, and consistency of the generated text. By incorporating this term, SynPO ensures a balanced optimization that simultaneously enhances preference alignment and sustains high-quality language generation.

4) *Hyperparameters:* The combination of the hyperparameter  $\alpha$  and the sigmoid function facilitates faster convergence during the early stages of preference optimization, while ensuring greater stability in later stages. The larger the value of  $\alpha$ , the more sensitive it is to the differences between positive and negative preferences. Therefore, its value is highly dependent on the quality of the preference dataset. The value of  $\beta$  determines the extent of influence of the language reward term on the joint optimization process. We recommend setting it to a relatively low value to encourage the model to maintain its language generation capabilities without overly affecting the preference optimization.

5) *Intuitive Interaction Between the Two Terms:* From an intuitive perspective, the two components of SynPO play complementary roles during optimization. The contrastive term, through the  $\exp(\log\text{-mean})$  structure, behaves like a *difficulty-aware discriminator*: it automatically assigns larger gradient signals to tokens with lower probabilities, which are often the semantically decisive or uncertain parts of a response. This encourages the model to focus on the critical tokens that distinguish positive and negative preferences, thereby strengthening preference alignment at the sentence level. In contrast, the language reward term acts as a *uniform stabilizer*.

TABLE I  
ABLATION STUDY ON THREE SCORING METRICS IN THE PIPELINE OF CONSTRUCTING PREFERENCE PAIRS

|                     | Criterion 1, 2, 3 | Criterion 1, 2 | Criterion 1, 3 | Criterion 2, 3 | Criterion 1 | Criterion 2 | Criterion 3 |
|---------------------|-------------------|----------------|----------------|----------------|-------------|-------------|-------------|
| Positive Preference | 2.26              | 2.21           | 2.17           | 2.20           | 2.15        | 2.09        | 2.08        |
| Negative Preference | 1.62              | 1.69           | 1.60           | 1.79           | 1.82        | 1.87        | 1.60        |

TABLE II  
ABLATION STUDY ON CONTRASTIVE DECODING (CD) AND THE SELF-RETROSPECTIVE STRATEGY

| Method             | Accuracy | Richness | Completeness | Fluency | Dynamics | Coherence | Average |
|--------------------|----------|----------|--------------|---------|----------|-----------|---------|
| Baseline           | 1.79     | 3.68     | 2.54         | 4.43    | 1.20     | 3.09      | 2.79    |
| CD only            | 1.91     | 3.66     | 2.61         | 4.44    | 1.18     | 3.08      | 2.81    |
| Retrospective only | 1.78     | 3.86     | 2.55         | 4.54    | 1.27     | 3.19      | 2.87    |
| CD & Retrospective | 1.90     | 3.85     | 2.59         | 4.52    | 1.28     | 3.17      | 2.88    |

By averaging token probabilities arithmetically, it distributes learning signals evenly across all tokens, encouraging the model to preserve overall fluency, grammaticality, and coherence. During training, the contrastive term drives the model to better separate preferred and dispreferred responses by emphasizing key tokens, while the language reward term prevents the optimization from over-focusing on a few difficult tokens and drifting away from general language modeling behavior. Their interaction results in a balanced optimization process that simultaneously sharpens preference discrimination and maintains strong generative language quality.

## V. EXPERIMENTS

### A. Constructing Long Video Caption Preference Pairs

1) *Effectiveness of Contrastive Decoding and Retrospective Strategy*: The impact of contrastive decoding and the self-retrospective strategy on the quality of generated captions is evaluated. We compare four settings: baseline (no enhancement), contrastive decoding only, self-retrospective only, and the combination of both. Each setting is applied to generate captions on the dataset of VDD [6], and the outputs are evaluated by an LLM along six dimensions: accuracy, richness, completeness, fluency, dynamics and coherence. As shown in Table II, combining both strategies achieves the highest scores across all metrics, outperforming either method alone. Specifically: (1) Contrastive decoding significantly improves accuracy (+6.7%) and completeness (+2.8%), indicating its effectiveness in reducing hallucinations and ensuring factual consistency; (2) The self-retrospective strategy excels in enhancing richness (+5.5%) and dynamics (+7.6%), demonstrating its ability to inject more detailed content; (3) The combined approach retains the strengths of both methods, achieving balanced improvements in all aspects.

2) *Ablation Study on Preference Pair Construction*: Figure 7 illustrates the performance of different augmentation methods across varying sampling counts. Using both techniques simultaneously achieves the best outcomes at identical sampling rates. It is notable that the self-retrospective strategy approximately doubles inference time, while contrastive decoding increases it by 50-75%. Given this trade-off, we find

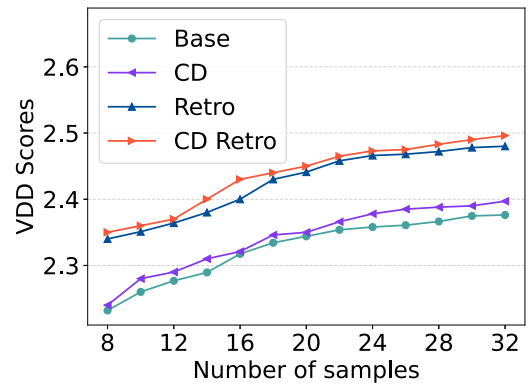


Fig. 7. VDD scores of positive preferences constructed via our pipeline under different sampling counts.

that employing the self-retrospective strategy with moderately increased sampling yields the most cost-effective approach for preference pair generation.

To assess the contribution of each of our three proposed scoring criteria to the quality of preference pairs, we conduct ablation studies using AuroraCap [15] as the base model, generating 10 samples per input with the self-retrospective strategy. Results for different criterion combinations are reported in Table I. The results show that all three criteria meaningfully contribute to final preference selection. Furthermore, Criterion 1 plays the most critical role in identifying high-quality positive preferences, whereas Criterion 3 demonstrates the strongest effectiveness in distinguishing negative preferences.

3) *Downstream Preference Learning*: To validate the effectiveness of the proposed pipeline, AuroraCap is fine-tuned using several methods, including SFT, DPO [9], and our SynPO. The preference pairs used for fine-tuning were generated from a subset of Sharegpt4video [34] through our automated pipeline, with additional details provided in Appendix A. Experiments are further conducted across diverse datasets and model configurations. The resulting models are evaluated on standard video captioning benchmarks (MSR-VTT [16], VATEX [17]) using CIDEr (C) [69] and METEOR (M) [70] metrics. In addition, we assess performance on VDC [15] and VDD [6] benchmarks, which employ

TABLE III

EXPERIMENTAL EVALUATION ON VARIOUS MODELS (AURORACAP [15], LLaVA1.6-7B-VIDEO [22], INTERVL2-8B [23]), DATASETS (SHAREGPT4VIDEO [34], CHARADES [61], PANDAS-70M [62]), AND FINE-TUNING APPROACHES (SFT, DPO, SYNPO AND OTHER VARIANTS). AS OBSERVED, ACROSS DIFFERENT MODELS (AURORACAP [15], LLaVA1.6-7B-VIDEO [22], INTERVL2-8B [23]) AND DATASETS (SHAREGPT4VIDEO [34], CHARADES [61], PANDAS-70M [62]), THE MODELS FINE-TUNED WITH SFT, DPO, AND SYNPO SIGNIFICANTLY OUTPERFORM THE BASELINE MODELS

| AuroraCap fine-tuned with different methods | VDC         |             |             |             |             | VDD         | Vatex       |             | MSRVTT      |             |
|---------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                                             | Camera      | Short       | Background  | Main Object | Detail      | Score       | CIDEr       | Meteor      | CIDEr       | Meteor      |
| AuroraCap (Base)                            | 1.22        | 1.79        | 1.58        | 1.45        | 1.70        | 2.00        | 38.4        | 18.6        | 33.2        | 10.8        |
| AuroraCap (SFT)                             | 1.43        | 1.85        | 1.73        | 1.72        | 1.91        | 2.18        | 39.2        | 19.0        | 34.0        | 11.2        |
| AuroraCap (DPO [9])                         | 1.39        | 1.89        | 1.70        | 1.73        | 1.94        | 2.23        | 39.6        | 18.9        | 34.1        | 11.4        |
| AuroraCap (DPO [12])                        | 1.55        | 1.88        | 1.78        | 1.79        | 1.96        | 2.30        | 41.1        | 19.3        | 34.8        | <b>11.5</b> |
| AuroraCap (IPO [63])                        | 1.44        | 1.83        | 1.74        | 1.76        | 1.95        | 2.21        | 39.8        | 19.1        | 34.3        | 11.3        |
| AuroraCap (KTO [64])                        | 1.53        | 1.88        | 1.73        | 1.78        | 1.97        | 2.27        | 40.1        | 19.0        | 34.3        | 11.4        |
| AuroraCap (CPO [65])                        | 1.42        | 1.81        | 1.75        | 1.72        | 1.94        | 2.25        | 40.8        | 18.9        | 34.1        | 11.2        |
| AuroraCap (SimPO [60])                      | 1.52        | 1.83        | 1.76        | 1.75        | 1.96        | 2.26        | 40.2        | 19.1        | 34.5        | 11.3        |
| AuroraCap (M-DPO [66])                      | 1.60        | 1.90        | 1.82        | 1.80        | 1.99        | 2.32        | 41.2        | 19.3        | 34.9        | 11.4        |
| AuroraCap (MrDPO [67])                      | 1.64        | 1.92        | 1.84        | 1.82        | 2.01        | 2.34        | 41.6        | 19.4        | 35.0        | 11.4        |
| AuroraCap (AuroraLong-2B [68])              | 1.49        | 1.82        | 1.76        | 1.77        | 1.95        | 2.25        | 40.8        | 19.0        | 34.3        | 11.2        |
| AuroraCap (SynPO-v1)                        | 1.72        | 1.89        | 1.87        | 1.83        | 2.02        | 2.35        | 42.1        | 19.5        | 35.2        | <b>11.5</b> |
| AuroraCap (SynPO-v2)                        | 1.74        | 1.93        | 1.88        | <b>1.87</b> | 2.03        | 2.37        | 42.3        | 19.5        | <b>35.4</b> | 11.4        |
| AuroraCap (SynPO-v3)                        | 1.75        | 1.91        | 1.90        | 1.84        | <b>2.05</b> | 2.38        | 42.3        | <b>19.6</b> | 35.3        | 11.3        |
| AuroraCap (SynPO-v4)                        | 1.48        | 1.87        | 1.80        | 1.75        | 1.98        | 2.25        | 40.9        | 19.2        | 34.6        | 11.2        |
| AuroraCap (SynPO-v5)                        | 1.57        | 1.78        | 1.82        | 1.77        | 1.97        | 2.29        | 41.2        | 19.3        | 34.5        | 11.2        |
| <b>AuroraCap (SynPO)</b>                    | <b>1.78</b> | <b>1.94</b> | <b>1.91</b> | <b>1.87</b> | 2.04        | <b>2.43</b> | <b>42.5</b> | <b>19.6</b> | <b>35.4</b> | <b>11.5</b> |
| <b>AuroraCap fine-tuned on more dataset</b> |             |             |             |             |             |             |             |             |             |             |
| Charades (SFT)                              | 1.40        | 1.87        | 1.68        | 1.75        | 1.90        | 2.21        | 39.5        | 19.2        | 33.8        | 11.3        |
| Charades (DPO)                              | 1.41        | 1.88        | 1.66        | 1.73        | 1.92        | 2.19        | 39.3        | 19.2        | 34.0        | 11.2        |
| <b>Charades (SynPO)</b>                     | 1.75        | <b>1.95</b> | 1.88        | 1.88        | 2.02        | 2.41        | 42.2        | 19.6        | <b>35.2</b> | 11.5        |
| Pandas-70M (SFT)                            | 1.45        | 1.86        | 1.71        | 1.74        | 1.92        | 2.22        | 39.8        | 19.2        | 34.1        | 11.2        |
| Pandas-70M (DPO)                            | 1.44        | 1.88        | 1.75        | 1.75        | 1.93        | 2.25        | 40.5        | 19.3        | 33.9        | 11.1        |
| <b>Pandas-70M (SynPO)</b>                   | <b>1.79</b> | 1.94        | <b>1.91</b> | <b>1.89</b> | <b>2.06</b> | <b>2.44</b> | <b>42.6</b> | <b>19.7</b> | 35.0        | <b>11.6</b> |
| <b>Models fine-tuned on Sharegpt4Video</b>  |             |             |             |             |             |             |             |             |             |             |
| LLaVA1.6-7B-video                           | 1.14        | 1.75        | 1.63        | 1.41        | 1.63        | 1.89        | 33.5        | 16.8        | 29.6        | 10.1        |
| LLaVA1.6-7B-video (SFT)                     | 1.37        | 1.83        | 1.72        | 1.66        | 1.88        | 2.13        | 34.7        | 17.3        | 30.8        | 10.5        |
| LLaVA1.6-7B-video (DPO)                     | 1.45        | 1.87        | 1.71        | 1.72        | 1.90        | 2.19        | 35.3        | 17.3        | 30.7        | 10.5        |
| <b>LLaVA1.6-7B-video (SynPO)</b>            | 1.74        | 1.90        | 1.94        | 1.85        | 2.00        | 2.36        | 37.3        | 18.1        | 32.1        | 10.9        |
| InternVL2-8B                                | 1.26        | 1.83        | 1.66        | 1.64        | 1.69        | 2.15        | 39.2        | 19.2        | 33.9        | 10.9        |
| InternVL2-8B (SFT)                          | 1.46        | 1.88        | 1.77        | 1.83        | 1.93        | 2.26        | 40.3        | 19.7        | 34.7        | 11.5        |
| InternVL2-8B (DPO)                          | 1.44        | 1.92        | 1.82        | 1.89        | 1.91        | 2.31        | 40.7        | 19.6        | 35.2        | 11.3        |
| <b>InternVL2-8B (SynPO)</b>                 | <b>1.80</b> | <b>1.96</b> | <b>1.95</b> | <b>1.97</b> | <b>2.04</b> | <b>2.48</b> | <b>42.8</b> | <b>20.1</b> | <b>36.3</b> | <b>11.7</b> |

LLM-based evaluation to better measure linguistic richness and factual consistency through their longer, richer reference captions(see Appendix A for details). As shown in Table III, all fine-tuned models significantly surpassed their baselines, confirming that our data construction pipeline yields high-quality data.

### B. Experiments of SynPO

1) *Comparison With DPO and Various Variants:* As shown in Table III, SynPO and several of its variants, as well as DPO variants, are compared under identical training settings and evaluation metrics (as defined in Section V-A.3). The results indicate that SynPO typically outperforms other variants, including those detailed in Table IV. Comparisons with various SynPO variants confirm that our modifications to the formula, specifically the incorporation of  $\sigma(\cdot)$ , logarithmic and exponential functions, yield significant performance improvements, validating the effectiveness and optimality of our approach.

3) *Training Recipe:* Training experiments are conducted using Llama3-8B [25] (Base and Instruct) and Mistral-7B [24] (Base and Instruct). For both Llama3-8B-Base and Mistral-7B-Base, we employ a training pipeline [73]. First, we train

a base model on the UltraChat-200k dataset [74] to obtain an SFT model. Then, we perform preference optimization on the UltraFeedback dataset [75] using the SFT model as the starting point. For Llama3-8B-Instruct and Mistral-7B-Instruct, we implement an on-policy evaluation strategy following SimPO [60]. Specifically, prompts from UltraFeedback are used to regenerate positive and negative preference pairs via SFT models. For each prompt, five responses are sampled from the SFT model and rank them using PairRM (LLM-Blender) [76]. The highest-ranked response is selected as the positive preference, and the lowest-ranked response is designated as the negative one.

4) *Evaluation:* Building upon recent advances in preference optimization [9], [73], we adopt a standard evaluation framework to assess model performance. We evaluate the general language capabilities of the models using the AlpacaEval 2 [71] and MT-Bench [72] benchmarks. The evaluation incorporates the following metrics: (1) win rate, which measures the model's effectiveness in responding to language queries; (2) length-controlled win rate, which evaluates the model's performance with respect to output length control; and (3) GPT-4 score, reflecting human preference judgments as scored by the GPT-4 model. As shown in Table V, our method consistently outperforms DPO and its variants across multiple

TABLE IV  
SYNPO VARIANTS FOR ABLATION STUDY ON THE LOG OPERATION, LANGUAGE TERM, ETC

| Method   | Objective Function                                                                                                                                                                                                                                                                                                                      |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SynPO-v1 | $\sigma \left( \alpha \cdot \exp \left( \frac{1}{ y_w } \sum_{t=1}^{ y_w } \log \pi_{\theta}(y_{w,t} x, y_{w,<t}) \right) - \alpha \cdot \exp \left( \frac{1}{ y_l } \sum_{t=1}^{ y_l } \log \pi_{\theta}(y_{l,t} x, y_{l,<t}) \right) \right)$                                                                                         |
| SynPO-v2 | $\sigma \left( \alpha \cdot \exp \left( \frac{1}{ y_w } \sum_{t=1}^{ y_w } \log \pi_{\theta}(y_{w,t} x, y_{w,<t}) \right) - \alpha \cdot \exp \left( \frac{1}{ y_l } \sum_{t=1}^{ y_l } \log \pi_{\theta}(y_{l,t} x, y_{l,<t}) \right) + \beta \cdot \frac{1}{ y_w } \sum_{t=1}^{ y_w } \log \pi_{\theta}(y_{w,t} x, y_{w,<t}) \right)$ |
| SynPO-v3 | $\sigma \left( \alpha \cdot \left( \frac{1}{ y_w } \sum_{t=1}^{ y_w } \log \pi_{\theta}(y_{w,t} x, y_{w,<t}) \right) - \alpha \cdot \left( \frac{1}{ y_l } \sum_{t=1}^{ y_l } \log \pi_{\theta}(y_{l,t} x, y_{l,<t}) \right) + \beta \cdot \frac{1}{ y_w } \sum_{t=1}^{ y_w } \pi_{\theta}(y_{w,t} x, y_{w,<t}) \right)$                |
| SynPO-v4 | $\sigma \left( \alpha \cdot \left( \frac{1}{ y_w } \sum_{t=1}^{ y_w } \log \pi_{\theta}(y_{w,t} x, y_{w,<t}) \right) - \alpha \cdot \left( \frac{1}{ y_l } \sum_{t=1}^{ y_l } \log \pi_{\theta}(y_{l,t} x, y_{l,<t}) \right) + \beta \cdot \frac{1}{ y_w } \sum_{t=1}^{ y_w } \log \pi_{\theta}(y_{w,t} x, y_{w,<t}) \right)$           |
| SynPO-v5 | $\left( \alpha \cdot \exp \left( \frac{1}{ y_w } \sum_{t=1}^{ y_w } \log \pi_{\theta}(y_{w,t} x, y_{w,<t}) \right) - \alpha \cdot \exp \left( \frac{1}{ y_l } \sum_{t=1}^{ y_l } \log \pi_{\theta}(y_{l,t} x, y_{l,<t}) \right) + \beta \cdot \frac{1}{ y_w } \sum_{t=1}^{ y_w } \pi_{\theta}(y_{w,t} x, y_{w,<t}) \right)$             |

TABLE V  
ALPACAEval2 [71] AND MT-BENCH [72] RESULTS UNDER THE FOUR SETTINGS. LC AND WR DENOTE LENGTH-CONTROLLED WIN RATE AND RAW WIN RATE, RESPECTIVELY

| Method       | Mistral-7B-Base |             |            | Mistral-7B-Instruct |             |            | Llama3-8B-Base |             |            | Llama3-8B-Instruct |             |            |
|--------------|-----------------|-------------|------------|---------------------|-------------|------------|----------------|-------------|------------|--------------------|-------------|------------|
|              | AlpacaEval 2    |             | MT-Bench   | AlpacaEval 2        |             | MT-Bench   | AlpacaEval 2   |             | MT-Bench   | AlpacaEval 2       |             | MT-Bench   |
|              | LC (%)          | WR (%)      | GPT-4      | LC (%)              | WR (%)      | GPT-4      | LC (%)         | WR (%)      | GPT-4      | LC (%)             | WR (%)      | GPT-4      |
| SFT          | 8.4             | 6.2         | 6.3        | 17.1                | 14.7        | 7.5        | 6.2            | 4.6         | 6.6        | 26.0               | 25.3        | 8.1        |
| DPO [9]      | 15.1            | 12.5        | 7.3        | 26.8                | 24.9        | 7.6        | 18.2           | 15.5        | 7.7        | 40.3               | 37.9        | 8.0        |
| DPOP [12]    | 16.1            | 12.8        | 7.4        | 27.1                | 24.6        | 7.7        | 16.7           | 14.3        | 7.6        | 45.2               | 39.1        | 8.2        |
| IPO [63]     | 11.8            | 9.4         | 7.2        | 20.3                | 20.3        | 7.8        | 14.4           | 14.2        | 7.4        | 35.6               | 35.6        | <b>8.3</b> |
| KTO [64]     | 13.1            | 9.1         | 7.0        | 24.5                | 23.6        | 7.7        | 14.2           | 12.4        | <b>7.8</b> | 33.1               | 31.8        | 8.2        |
| CPO [65]     | 9.8             | 8.9         | 6.8        | 23.8                | 28.8        | 7.5        | 10.8           | 8.1         | 7.4        | 28.9               | 32.2        | 8.0        |
| SimPO [60]   | 21.5            | 20.8        | 7.3        | 32.1                | 34.8        | 7.6        | 22.0           | 20.3        | 7.7        | 44.7               | 40.5        | 8.0        |
| <b>SynPO</b> | <b>22.9</b>     | <b>22.1</b> | <b>7.7</b> | <b>37.9</b>         | <b>39.8</b> | <b>7.9</b> | <b>25.7</b>    | <b>23.1</b> | <b>7.7</b> | <b>49.0</b>        | <b>46.2</b> | <b>8.3</b> |

models on these metrics. These results demonstrate both the effectiveness and broad applicability of SynPO.

## VI. CONCLUSION

This research addresses two fundamental challenges hindering fine-grained video captioning: the lack of scalable, high-quality preference data and the practical limitations of standard DPO. To generate preference data efficiently, we develop an automated pipeline requiring neither human annotation nor access to stronger VLMs. Concurrently, our theoretical and empirical analysis reveals DPO’s core issues: excessive focus on negative examples and deviation from ranking optimization. Our solution, SynPO, counteracts these by rebalancing preference signals and incorporating generation-preserving terms, leading to improved language capability and training efficiency. SynPO’s effectiveness and broad applicability are validated through extensive experiments on diverse video captioning and NLP benchmarks, where it consistently outperforms existing methods.

## LIMITATIONS

While SynPO effectively addresses the optimization objective shift in preference alignment for video-language models, it possesses several limitations that warrant further exploration:

- **Performance in Deterministic Reasoning:** Our observations suggest that while SynPO excels in enhancing the fluency and descriptive quality of video captions, its relative gains may be less pronounced in highly deterministic domains such as mathematical reasoning or code generation within video contexts. In these areas, iterative reinforcement learning methods with explicit reward modeling (e.g., GRPO and its variants) might offer more precise optimization trajectories compared to the current contrastive approach.
- **Generalizability to Broader Video Domains:** Although SynPO demonstrates robust results in standard video captioning tasks, its application to the “broader video domain”—such as ultra-long videos or those requiring fine-grained temporal action localization—remains to be fully validated. The current objective primarily focuses on sequence-level coherence and quality. For tasks requiring extreme temporal precision, the uniform contribution of the language reward term might need further adaptation to account for the varying importance of specific temporal segments.

## APPENDIX A

### IMPLEMENTATION DETAILS FOR VIDEO CAPTIONING

In Table III, we conduct ablation studies to evaluate the effectiveness of our data generation approach and SynPO.

**Algorithm 1** SynPO Training Step**Require:** Preference dataset  $\mathcal{D}$ , model  $\pi_\theta$ , hyperparameters $\alpha, \beta$ 1: **for** each batch  $(x, y_w, y_l) \sim \mathcal{D}$  **do**

2:   Compute token probabilities for positive sequence:

$$p_t^w = \pi_\theta(y_{w,t} \mid x, y_{w,<t}), \quad t = 1, \dots, |y_w|$$

3:   Compute token probabilities for negative sequence:

$$p_t^l = \pi_\theta(y_{l,t} \mid x, y_{l,<t}), \quad t = 1, \dots, |y_l|$$

4:   Compute sequence-level geometric means:

$$G_w = \exp \left( \frac{1}{|y_w|} \sum_{t=1}^{|y_w|} \log p_t^w \right)$$

$$G_l = \exp \left( \frac{1}{|y_l|} \sum_{t=1}^{|y_l|} \log p_t^l \right)$$

5:   Compute language reward term:

$$R_w = \frac{1}{|y_w|} \sum_{t=1}^{|y_w|} p_t^w$$

6:   Compute SynPO preference score:

$$s = \alpha(G_w - G_l) + \beta R_w$$

7:   Compute loss:

$$\mathcal{L} = -\log \sigma(s)$$

8:   Update model parameters  $\theta$  using  $\nabla_\theta \mathcal{L}$ 9: **end for**

In the first experiment, we fine-tune AuroraCap using DPO and its various variants. The fine-tuning dataset is generated using our proposed data generation pipeline on a subset of ShareGPT4Video. In the second part of Table III, we only change the source dataset used in the data generation pipeline to verify the general adaptability of our pipeline and SynPO. In the third part of Table III, we use the same preference pairs generated from ShareGPT4Video as in the first experiment to fine-tune several popular multimodal models, and evaluate the resulting models.

**A. Data Generation Setup**

For the video detailed captioning experiment evaluating the effectiveness of our approach (Table II), we sample over 10,000 source videos from the ShareGPT4Video dataset. We employ a sampling strategy with a count of 10 per video, using temperature = 0.9, top\_p = 0.95, and top\_k = 32 to generate diverse candidate captions. The LLM used to score all generated captions is Qwen-Plus-2025-01-25. Among the candidates, the caption receiving the highest score is selected as the positive preference, while the one with the lowest score is treated as the negative preference.

**B. Training Setup**

The maximum number of training epochs is set to 5, and the best-performing model based on validation performance is selected for final evaluation. We use the AdamW optimizer with a linear learning rate scheduler incorporating warmup. The warmup ratio is set to 0.1, and the learning rate is  $5 \times 10^{-6}$ . The batch size is fixed at 32 during training.

**C. Evaluation Setup**

During inference, we adopt a greedy decoding strategy for caption generation. Additionally, four widely-used benchmark datasets are employed to comprehensively evaluate the model's performance across multiple dimensions.

## APPENDIX B

### PSEUDOCODE FOR SYNPO

See Algorithm 1.

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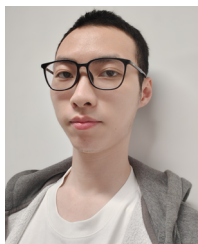


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